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https://orcid.org/0000-0002-5834-0812

Davide Conficconi, Ph.D.

EMPLOYMENT

NOW
MAR 2024
Untenured Track Assistant Professor, Politecnico di Milano, Milan, Italy
Established research line in (co-)designing of Domain-Specific Systems and Architectures at any scale
Investigating energy efficient and neuromorphic computing systems
Achieved interdisciplinary co-design activities with biomedical researchers on efficient medical image registration system
Authored a patent pending on novel virtual memory method and system, possibly shifting the paradigm
Authored the proposal and led a work package EIC pathfinder Cancerscan: accelerated digital twin for pancreatic tumor
Achieved interdisciplinary co-design activities with space researchers on autonomous deep space satellites guidance
Exploring interdisciplinary co-design activities with diverse researchers on accelerating the whiskey aging production

System Administrator of Computer Architectures NECSTLab machines (racks, towers, embedded systems)
Promoted Politecnico di Milano to the wider public at the Open day
Developed myself through the Politecnico di Milano talent development program, soft-skills courses, and workshops

FEB 2024
MAR 2022
Post-Doctoral Researcher, Politecnico di Milano, Milan, Italy
Established the lab research line on Quantum Error Correction with FPGAs
Amplified my research line on Domain-Specific Architectures (Compiler, ISA, Microarchitecture) for Regular Expressions
Coordinated the general FPGA research line in the NECSTLab
System Administrator of Computer Architectures NECSTLab machines (racks, towers, embedded systems)
Promoted Politecnico di Milano to the wider public at the Open day

FEB 2022
SEPT 2021
Research intern, IBM Research Zurich, Zurich, Switzerland
Designed a memory mapped characterizer, stressor and fault detection for network-attached cloudFPGA system
Created a design methdology for network-attached domain-specific accelerators for the hybrid multi-cloud era, cFp_Zoo

FEB 2019
AUG 2018
Research intern, Xilinx (now AMD) Research, Dublin, Ireland
Designed an accelerator for bit-serial data layout transformation, published in ACM TRETS'19
Created a bit-serial neural network inference acceleration pipeline in BISMO

JUN 2018
JAN 2018
Research Assistant, Oracle Labs, Zurich, Remote from Italy
Developed graph and data visualization applications exploration with internal labs tool

JUL 2018
JAN 2018
Research intern, UniCredit R&D, Milan, Italy
Explored financial application targeting FPGAs

QUALIFICATIONS

NOV 2024
Associate Professor (equivalent to Reader) **Italian qualification**
Information Processing Systems, IINF-05/A; ING-INF/05;

FEB. 2022
NOV 2018
Ph.D. Information Technology: **Computer Science and Engineering**
Politecnico di Milano
Outlined a clear taxonomy of Domain-Specific Architectures based on Reconfigurable Systems
Designed a taxonomy and historical perspectives on how to program FPGAs with HDL, HLS, DSL
Developed two classes (speculative and enumerative) of Domain-Specific Architectures for Regex matching
Developed a configurable FPGA accelerator for Mutual Information, essential for Biomedical Cancer Image Analysis
Promoted Politecnico di Milano to the wider public at the Open day

JULY 2018
OCT 2015
M.Sc. Computer Science and Engineering
Politecnico di Milano
Developed a scalable multi-core FPGA domain-specific architecture and compiler for Regular Expression matching
Developed a methodology to integrate Chisel accelerators on PCIe FPGA cards with SDAccel
Promoted Politecnico di Milano to the wider public at the Open day
Scholarship for High Merits, partial tuition waiver

SEPT 2015
SEPT 2012
B.Sc. Computer Science and Engineering
Politecnico di Milano

GRANTS AND FUNDING

Total as (Co-)PI: €45K; as key personnel (proposal writer, researcher, work package leader): €865K

JAN 2026 **Informatics-Huawei distinguished visitor scheme** funds (€20K), joint research visiting at University of Edinburgh on "*Exploring Emerging Scalable/Composable Compute and Memory Interconnects for Sustainable Energy Efficient Computing*"
PIs: Davide Conficconi and Prof. Antonio Barbalace (University of Edinburgh).

SEPT 2025 **TUM Global Incentive Fund (GIF)** seed funds (€10K) on "*WHISKEY-AI: Accelerating Food Fermentation and Beverage Aging with Physics-Informed Edge AI to Reduce Environmental Impact*"
a collaboration among researchers from TUM, Imperial, and Polimi thanks to ETA fellowship.

AUG 2025 **EIC** (European Innovation Council) **Pathfinder** "***CANCERSCAN**: Smart pathology slide scanner for diagnosis and patient-specific treatment recommendation in oncology*"
Duration 36 months. Total funds ~€3M. Politecnico di Milano funds:~€588K. **PI**: M. D. Santambrogio
Proposal writer, **Work Package Leader** of WP4 "*Hardware Optimization & Embedding*"

FEB 2025 **AMD Fund for Academic Research (FAR) Grant** (€20K) and hardware donations (€15K)
PI: Davide Conficconi, **Topic**: Domain-Specific Heterogeneous Systems research and teaching activities.

JAN 2024 **ERC** (European Research Council) **Proof of Concept** "***GUIDO**: A Computing Unit for Autonomous Spacecraft Guidance in Deep Space*"
Duration 12 months. Politecnico di Milano funds ~€150K. DEIB funds:~€25K. **PI**: F. Topputo
Work Package Leader of WP3 "*Simulation Execution*"

JAN 2023 **Huawei Technologies Zurich Research Center Grant** (~€250K)
FEB 2022 **Proposal writer**, **Post-Doc Researcher**, **Topic**: Templated Spatial Architectures. **PI**: M. D. Santambrogio.

SEPT 2022 **NVIDIA Academic Hardware Grant Program** Bluefield DPU 2 and 2x NVIDIA A5000 (~€10K)
SEPT 2021 **Proposal writer**, **Researcher**. **Topic**: SW-programmable Domain Specific Architectures for Regex. **PI**: M. D. Santambrogio.

TEACHING ACTIVITIES

DEC 2025 **Graduate Course**: *"Advanced Topics in Codesign of Domain-Specific Accelerated Computing Architectures and Systems"*
OCT 2025 Course organizer, Lecturer, and Marker. Lectures from Prof. L. Josipovic (ETH Zurich), M. Langhammer (Altera)
Topics: codesign methodologies, domain-specific computing, conference simulation process.
CFU 5. **Participants**: 19. **Students Evaluation**: Not Yet Avaialable

FEB 2026 **Graduate Course**: *"Advanced Computer Architectures"*, Computer Science and Engineering, Politecnico di Milano
FEB 2020 Lecturer, Tutor, Marker. Course Organizers: M. Santambrogio (20, 21), D. Sciuto (22), Myself (23), C. Pilato (24, 25, 26)
Topics: CPU, Performance, ILP, TLP, DLP, Heterogeneous Computing.
CFU 5. **Average Participants**: 200. **Students Evaluation**: High

JAN 2026 **Undergraduate Course**: *"Computer Science for Data Visualization"*, Communication Design, Politecnico di Milano
SEPT 2023 Course Organizer, Lecturer, Marker.
Topics: Basics Informatics JS, p5JS for Dataviz. **CFU** 5. **Average Participants**: 60. **Students Evaluation**: High

JAN 2024 **Undergraduate Course**: *"Basics of Informatics"*, Biomedical Engineering, Politecnico di Milano
FEB 2023 Course organizer, Lecturer, Marker.
CFU 5. **Participants**: 191. **Students Evaluation**: Average

FEB 2024 **Innovative Course**: *"FPGA101: From Reconfigurable to Domain-Specific Systems"*, Any degree in Politecnico di Milano
OCT 2024 Course Organizer, Lecturer, Tutor. Editions: sp'25, fall'24.
Topics: FPGA design flows and system methodologies, NPU domain-specific architectures.
CFU 5. **Average Participants**: 17. **Students Evaluation**: High

FEB 2024 **Innovative Course**: *"FPGA101"*, Any degree in Politecnico di Milano
FEB 2020 Lecturer, Tutor. Course Organizer: M. Santambrogio. Editions: sp'24, fall'23, sp'23, fall'22, sp'22, sp'21, sp'20.
Topics: FPGA design flows and system design methodologies. **CFU** 5. **Average Participants**: 15.

JULY 2022 **Undergraduate Course**: *"Hardware and Accelerators: FPGAs for AI"*, Specialization degree
Course organizer, Lecturer. CEFRIEL–Politecnico di Milano
CFU 5. **Participants**: 20.

SEPT 2023 **Summer School**: *"Creative Lab"*, Cyber-Physical Systems (CPS) Summer School
MAY 2023 Course Organizer, Lecturer, Tutor.
Hackathon-like activities with Kria KV260.**Participants**: 30.

FEB 2020 FEB 2019 **Graduate Course**: *"Digital Systems and Design Methodologies 1"*, Computer Science and Electronics Eng., Politecnico di Milano
Lecturer, Teaching Assistant, Marker. Course Organizers: F. Ferrandi.
Topics: Hardware design flow, ASIC vs FPGA, Logic Synthesis, Timing Analysis, High-level Synthesis, VHDL IP Modeling.
CFU 5. **Average Participants**: 80. **Students Evaluation**: Medium

FEB 2020 FEB 2019 **Undermgraduate Course**: *"Final Projects of Digital Design"*, Computer Science and Engineering, Politecnico di Milano
Project Tutor. Course Organizers: G. Palermo)
Topics: Digital Circuits Modelling, Combinational vs Sequential design, Datapath and Control Logic, Memory communication.
CFU 1. **Average Participants**: 147. **Students Evaluation**: not available.

SUPERVISION ACTIVITIES

PhD Students:

- Main Advisor of 2 PhD student in Computer Science and Engineering: day-to-day supervision, supporting methodology learning, supporting writing, career guidance, mentoring
- Co-Advisor of 1 PhD student (50%) interdisciplinary Aerospace Systems and Computer Science and Engineering: supporting methodology learning on HW/SW co-design, mentoring
- Supporting 3 PhD students in achieving Computer Science and Engineering.
  Demonstrated being present in all their publications thanks to which they obtained their PhD degree.

Graduate Thesis:

- Main Advisor of 9 Master students in Computer Science and Engineering (Politecnico di Milano) 1 Master student in Computer Science (University of Illinois Chicago), 9 project proposer and project direction advisor, 1 project direction advisor
- Co-Advisor of 5 Master students in Computer Science and Engineering, 4 project proposer and tutor, 1 project direction advisor
- Results in 5 Publications (SIGMETRICS'26, CGO'24, DAC'24, TECS'23, ISCAS'23)

Graduate and Undergraduate Research Projects:

- Supervising 45 research projects of any students grade, mainly Computer Science and Engineering, providing initial idea, ideas and discussion throughout the project, guiding their work, and providing computation infrastructure
- Part of these projects resulted in 23 Publications, partially overlapped with the thesis outcome
- One at the intersection with computer security resulted in CVE-2025-63094, Spectre on XiangShan RISC-V high-performance CPU and a publication [C5]

INVITED TALKS

2024 "Accelerating Iterative Rigid Medical Image Registration via Domain-Specialization: from Embedded to High-Performance Systems" at AMD-Xilinx, June 2024, San Jose, California (US)
"Exploring Domain-Specialization in the Regular Expression Field", at AMD-Xilinx June 2024, San Jose, California (US)
"From FPGA To AI Engine: Beyond Mutual Information Limits", at AMD Open Hardware Competition '24

2023 "Domain-Specific Computing Research Line at NECSTLab", at Boston University, Boston, MA (USA).
"Faber: Hardware/Software Toolchain for Image Registration", at Northeastern University, Boston, MA (USA).

2022 "Domain-Specific Computing Research Line at NECSTLab", at Huawei Research at Polimi, Milan (ITA)
"Faber: Hardware/Software Toolchain for Image Registration", at Northwestern University and UIC, IL (USA).
"Software-programmable Domain-Specific Architectures for Regular Expressions", virtually held to NVIDIA DPU team.

2019 "DRACO: Domain-specific Reconfigurable Architecture Computer Organization",
Open Networking Foundation (ONF), Menlo Park, CA (USA).
Lawrence Berkeley National Laboratory, Berkeley, CA (USA)
Xilinx, San Josè, CA (USA)

2018 "TiReX: Tiled Regular eXpression matching architecture", Xilinx, San Josè, CA (USA).

2017 "TiReX: Tiled Regular eXpression matching architecture", Microsoft Research in PoliMi, Milan (ITA)
"TiReX: Tiled Regular eXpression matching architecture", Xilinx, San Josè, CA (USA).

SELECTED AWARDS OF LAST FIVE YEARS

2025 **European Talent Academy fellowship**, selected among the best 7 researchers from Polimi.
Theme: "*Water and Food for Healthy and Resilient Societies*".
Training programme among Politecnico di Milano, Imperial, and TUM for joint proposals.

25-23 **IEEE Artifacts** *of Available, Reusable, Validated & Reproduced* for [C13], RAW24a, RAW23

25-21 **ACM Artifacts** *of Available, Reusable, Validated & Reproduced* for [C11], CGO24, [C34]

2024 **Supervisor of a Winner Team of AMD OHW Contest**: *From FPGA To AI Engine: Beyond Mutual Information Limits*
G. Brunetta, F. Santambrogio

2024 **Interdisciplinary Ph.D. Scholarship funds** for a *joint program* among *DAER-DEIB* departments of Politecnico di Milano.

2024 **HiPEAC Paper award** for [C16]

Best Poster award for RAW24b

2023 **HiPEAC Summer School ACACES Grant**, full waiver

2023 **Young People Programme** award at DATE'23

2023 **Co-Supervisor of a Winner Team of AMD OHW Contest**, *Heterogeneous Highly Integrated Systems for Image Registration*,
G. Sorrentino, P. Galfano, ICCD24,[C13]

2022 **Open Research Objects** research artifact available open-source for IISWC22

2022 **ASPLOS 2022 Student Travel Award**

2022 **Co-Supervisor of a Winner Team of XOHW Design Contest** *A Journey to the center of the 3D Space*,
G. Sorrentino, M. Venere, CODES/TECS23

2022 **IBM Best PhD Intern Presentation** at Cloud & AI Systems Research (CAISR) department Intern Symposium July '22

2022 **First Place IEEE 2022 Lance Stafford Larson Award** on [C34]

RESEARCH COMMUNITY ACTIVITIES

Conference Committee and Organization

PROGRAM CHAIR	RAW '26 RAW '25
ARTIFACT EVALUATION CHAIR	FCCM '26, FPL '26 RAW 24, 23
PROGRAM COMMITTEE	CGO '26, FCCM '26, FPL '26, Shadow EuroSys'26, DATE '26, ISLPED 2026, HEART'26, ASAP '26, GLSVLSI'26 FCCM '25, FPL '25,HEART'25, SAC'25 - EMBS Track, GLSVLSI '25, RAW '25, HPCC '25, EUROCON '25 HPCC '24, PACRIM '24 FPL '24, FCCM '24, GLSVLSI '24, HEART '24, SAC'24 - EMBS Track , RAW '24 ISPA '23, GLSVLSI '23, RAW '23 RAW '22
ARTIFACT EVALUATION COMMITTEE	ASPLOS '26 ASPLOS '25, CGO '25 CGO '24 MICRO '23, ASPLOS '23, CGO '23, PLDI '23 ASPLOS '22
TUTORIAL ORGANIZER	"Faber: a Hardware/Software Toolchain for Image Registration" at EUROCON '23 Chisel Bootcamp at ASAP '18

Journal Activities

GUEST EDITOR	Special Issue on top the papers from the Reconfigurable Architecture Workshop (RAW) 2025 ACM Transactions on Reconfigurable Technology and Systems (TRETS)
GUEST EDITOR	Special Issue on Secure and Efficient Distributed Computation for Emerging Systems on the Edge (SUNRISE) 2024 Elsevier Journal of Parallel and Distributed Computing (JPDC)

REVIEWER	Revising 60+ manuscripts across different journals
ACM	TRETS (25-23), TODAES (25,21), TECS (23,22), TACO (24)
IEEE	TCAD (25-24,22), CAL (25), TPDS (24,21,20) TVLSI (24,23), TETC (24,23), Access (24,22,21),ESL (24,23)
ELSEVIER	COMNET (24), SoftwareX (24), COSE (21), MICPRO (21)

PROFESSIONAL MEMBERSHIPS

IEEE	Member (2017)
ACM	Member (2020)
HIPEAC	Affiliated Member (2023)
IEEE	Computer Society (2017)
ACM	SIGARCH Member (2024)
IEEE	Council on Electronic Design Automation (2024)
IEEE	CS Technical Community on Computer Architecture
IEEE	Larson Award Chair (2025-2024)

SCIENTIFIC PRODUCTIVITY AND IMPACT

Full list of publications available at this link.

- (Co-)Author of **38** peer-reviewed conferences publications such as SIGMETRICS, FCCM, DAC, CGO, FPGA, EUROSYS;
- (Co-)Author of **13** journal papers including ACM TRETS, CSUR, IEEE TPDS, TETC, JBHI;
- (Co-)Author of **1** Book Chapter.

Research Impact:

- Atuhored a patent pending and filed: "Method and System of Virtual Memory Management", number 102025000029050, possibly revolutionizing the current approach
- Obtained AMD to support my research and teaching activities with hardware donations and money fundings
- Achieved interdisciplinary and international grant with TUM, Imperial, and Polimi partners
- Ph.D.-level course held with Prof. Lana Josipovic (ETH Zurich) and Martin Langhammer (Altera)
- Achieved Publications joint with international univeristies (University of Edinburgh, ETH Zurich, MIT) and research institutes (Huawei, Riken, Xilinx, AMD)
- Achieved the first publication at FPGA conference [C34] for an FPGA lab where I was doing my PhD
- Coordinated one of the most complex design, to date, leveraging the full heterogeneity of a Versal AIE-FPGA system [C13]
- [C11] has been subject of discussion with a Google researcher at CGO'25
- Discussed with international institutions on collaborating for a regex engine thanks to [J11]
- Authored 25 open-source artifacts, 7 of these also awarded for reproducible research with the corresponding publications

Quantitative Metrics:

Based on Google Scholar:	h-index **11**	citations **470**
Based on Scopus:	h-index **9**	citations **276**

Journal Publications

[J13] Francesco Peverelli, Daniele Paletti, and **Davide Conficconi**. "DFlows: A Flow-based Programming Approach for a Polyglot Design-Space Exploration Framework". In: *ACM Transactions on Reconfigurable Technology and Systems (TRETS)* 1.1 (2025), pp. 1–32. DOI: 10.1145/3717837.

[J12] Federico Valentino, Beatrice Branchini, **Davide Conficconi**, Donatella Sciuto, and Marco D. Santambrogio. "QUEKUF: an FPGA Union Find Decoder for Quantum Error Correction on the Toric Code". In: *ACM Transactions on Reconfigurable Technology and Systems (TRETS)* 1.1 (2025), pp. 1–32. DOI: 10.1145/3733239.

[J11] Marco Venere, Beatrice Branchini, **Davide Conficconi**, Donatella Sciuto, and Marco D. Santambrogio. "Rock the QASBA: Quantum Error Correction Acceleration via the Sparse Blossom Algorithm on FPGAs". In: *ACM Transactions on Reconfigurable Technology and Systems (TRETS)* 1.1 (2025), pp. 1–32. DOI: 10.1145/3723168.

[J10] Raffaele Berzoini, Eleonora D'Arnese, **Davide Conficconi**, and Marco D. Santambrogio. "NERONE: the Fast Way to Efficiently Execute Your Deep Learning Algorithm at the Edge". In: *IEEE Journal of Biomedical and Health Informatics (J-BHI)* (2023), pp. 1–9. DOI: 10.1109/JBHI.2023.3296142.

[J9] Emanuele Del Sozzo, **Davide Conficconi**, and Kentaro Sano. "Across Time and Space: Senju's Approach for Scaling Iterative Stencil Loop Accelerators on Single and Multiple FPGAs". In: *ACM Transactions on Reconfigurable Technology and Systems (TRETS)* (2023). DOI: 10.1145/3634920.

[J8] Giuseppe Sorrentino, Marco Venere, **Davide Conficconi**, Eleonora D'Arnese, and Marco D Santambrogio. "HEPHAESTUS: Codesigning and Automating 3D Image Registration on Reconfigurable Architectures". In: *ACM Transactions on Embedded Computing Systems (TECS)* 22.5s (2023). ISSN: 1539-9087. DOI: 10.1145/3607928.

[J7] Alberto Zeni, Emanuele Del Sozzo, Eleonora D'Arnese, **Davide Conficconi**, and Marco D Santambrogio. "Starlight: A Kernel Optimizer for GPU Processing". In: *Journal of Parallel and Distributed Computing* (2023). DOI: 10.1016/j.jpdc.2023.104832.

[J6] **Davide Conficconi**, Emanuele Del Sozzo, Filippo Carloni, Alessandro Comodi, Alberto Scolari, and Marco Domenico Santambrogio. "An Energy-Efficient Domain-Specific Architecture for Regular Expressions". In: *IEEE Transactions on Emerging Topics in Computing* (2022). DOI: 10.1109/TETC.2022.3157948.

[J5] Eleonora D'Arnese, **Davide Conficconi**, Emanuele Del Sozzo, Luigi Fusco, Donatella Sciuto, and Marco D Santambrogio. "Faber: a Hardware/Software Toolchain for Image Registration". In: *IEEE Transactions on Parallel and Distributed Systems* (2022). DOI: 10.1109/TPDS.2022.3218898.

[J4] Emanuele Del Sozzo, **Davide Conficconi**, Alberto Zeni, Mirko Salaris, Donatella Sciuto, and Marco Domenico Santambrogio. "Pushing the Level of Abstraction of Digital System Design: a Survey on How to Program FPGAs". In: *ACM Computing Surveys (CSUR)* (2022). DOI: 10.1145/3532989.

[J3] Daniele Parravicini, **Davide Conficconi**, Emanuele Del Sozzo, Christian Pilato, and Marco D Santambrogio. "CICERO: A Domain-Specific Architecture for Efficient Regular Expression Matching". In: *ACM Transactions on Embedded Computing Systems (TECS)* 20.5s (2021), pp. 1–24. DOI: 10.1145/3476982.

[J2] Enrico Reggiani, Emanuele Del Sozzo, **Davide Conficconi**, Giuseppe Natale, Carlo Moroni, and Marco D Santambrogio. "Enhancing the scalability of multi-fpga stencil computations via highly optimized hdl components". In: *ACM Transactions on Reconfigurable Technology and Systems (TRETS)* 14.3 (2021), pp. 1–33. DOI: 10.1145/3532989.

[J1] Yaman Umuroglu, **Davide Conficconi**, Lahiru Rasnayake, Thomas B Preusser, and Magnus Själander. "Optimizing bit-serial matrix multiplication for reconfigurable computing". In: *ACM Transactions on Reconfigurable Technology and Systems (TRETS)* 12.3 (2019), pp. 1–24. DOI: 10.1145/3337929.

Conference Publications

[C38] Claudio Di Salvo, Emanuele Del Sozzo, Giuseppe Sorrentino, Eleonora D'Arnese, Paolo Panicucci, and **Davide Conficconi**. "Are We Ready to Enable Satellite Autonomy Through on Board Image Registration?" In: *Proceedings of the ACM on Measurement and Analysis of Computing Systems*. 2026, pp. 1–20. DOI: accepted--to--appear.

[C37] Valentino Guerrini, Giuseppe Sorrentino, Alessandro Barenghi, and **Davide Conficconi**. "ReFHE-NTT: Resource-Driven NTT FPGA Architecture for Fully Homomorphic Encryption". In: *2026 IEEE Symposium on Field-Programmable Custom Computing Machines (FCCM)*. 2026, pp. 1–12. DOI: accepted--to--appear.

[C36] Valentino Guerrini, Giuseppe Sorrentino, and **Davide Conficconi**. "Exploring Resource-Efficient NTT FPGA Accelerator for Fully Homomorphic Encryption". In: *2026 Design, Automation & Test in Europe Conference (DATE)*. IEEE. 2026, pp. 1–2. DOI: accepted--to--appear.

[C35] Giulio Mantovi, Davide Paltrinieri, Giuseppe Sorrentino, Christian Pilato, and **Davide Conficconi**. "A Power-Centric Methodology to Characterize Edge AI SoC with Limited Telemetry Capabilities". In: *2026 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*. IEEE. 2026, pp. 1–8. DOI: accepted--to--appear.

[C34] Rey Paraula, Roberto Petenzi, Alessandro Bertani, and **Davide Conficconi**. "How Secure is a High-Performance RISC-V Core? A Spectre V1 Case Study on XiangShan Open-Source CPU". In: *Proceedings of the 19th European Workshop on Systems Security*. 2026, pp. 1–6. DOI: accepted--to--appear.

[C33] Giuseppe Sorrentino, Paolo S. Galfano, Claudio Di Salvo, Eleonora D'Arnese, and **Davide Conficconi**. "Adaptive AIE–PL Systems for Efficient End-to-End Pyramidal 3D Image Registration". In: *2026 IEEE Symposium on Field-Programmable Custom Computing Machines (FCCM)*. 2026, pp. 1–12. DOI: accepted--to--appear.

[C32] Marco Venere, Giuseppe Sorrentino, Benjamin Ramhorst, Maximilian Jakob Heer, Lucian Petrica, Dario Korolija, Marco Domenico Santambrogio, **Davide Conficconi**, Gustavo Alonso, and Kenneth O'Brien. "RoPeerTo: A Datacenter-Scale Architecture for Peer-To-Peer DMA between GPUs and FPGA". In: *Proceedings of the Twentyfirst European Conference on Computer Systems*. EuroSys '26. ACM. 2026, pp. 1–20. DOI: accepted--to--appear.

[C31] Eleonora Cabai, Giuseppe Sorrentino, Marco D. Santambrogio, and **Davide Conficconi**. "Accelerating K-Means: A Vectorized Approach for AI Engines & Neural Processing Units". In: *2025 35th International Conference on Field-Programmable Logic and Applications (FPL)*. IEEE. 2025, pp. 1–5. DOI: 10.1109/FPL68686.2025.00053.

[C30] Eleonora Cabai, Giuseppe Sorrentino, Marco Domenico Santambrogio, and **Davide Conficconi**. "A Hardware/Software Co-Design Approach for Versal-Based K-means Acceleration". In: *2025 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*. IEEE. 2025, pp. 1218–1218. DOI: 10.1109/IPDPSW66978.2025.00196.

[C29] Federico Mansutti, Davide Ettori, Giuseppe Sorrentino, Marco Domenico Santambrogio, and **Davide Conficconi**. "Towards a Methodology to Leverage Alveo Versal System Usability And Parallelization". In: *2025 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*. IEEE. 2025, pp. 1219–1219. DOI: 10.1109/IPDPSW66978.2025.00197.

[C28] Andrea Somaini, Filippo Carloni, Giovanni Agosta, Marco D Santambrogio, and **Davide Conficconi**. "Combining MLIR Dialects with Domain-Specific Architecture for Efficient Regular Expression Matching". In: *IEEE/ACM International Symposium on Code Generation and Optimization*. 2025. DOI: 10.1145/3696443.3708916.

[C27] Giuseppe Sorrentino and Davide Conficconi. "Towards Accelerated Healthcare Federated System Through Heterogeneous Accelerators". In: *2025 35th International Conference on Field-Programmable Logic and Applications (FPL)*. 2025, pp. 1–2. DOI: 10.1109/FPL68686.2025.00055.

[C26] Giuseppe Sorrentino, Paolo S. Galfano, Eleonora D'Arnese, and **Davide Conficconi**. "Soaring with TRILLI: an HW/SW Heterogeneous Accelerator for Multi-Modal Image Registration". In: *2025 IEEE Symposium on Field-Programmable Custom Computing Machines (FCCM)*. 2025, pp. 1–12. DOI: 10.1109/FCCM62733.2025.00040.

[C25] Giuseppe Sorrentino, Paolo S. Galfano, Eleonora D'Arnese, and **Davide Conficconi**. "VOTED – Versal Optimization Toolkit for Education and Heterogeneous Systems Development". In: *2025 IEEE International Symposium on Circuits and Systems (ISCAS)*. 2025, pp. 1–5. DOI: 10.1109/ISCAS56072.2025.11043842.

[C24] Roberto Alessandro Bertolini, Filippo Carloni, **Davide Conficconi**, and Marco D. Santambrogio. "POCA: a PYNQ Offloaded Cryptographic Accelerator on Embedded FPGA-based Systems". In: *IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*. 2024, p. 194. DOI: 10.1109/IPDPSW63119.2024.00054. URL: https://doi.org/10.1109/IPDPSW63119.2024.00054.

[C23] Filippo Carloni, **Davide Conficconi**, and Marco D Santambrogio. "ALVEARE: a Domain-Specific Framework for Regular Expressions". In: *61st ACM/IEEE Design Automation Conference (DAC '24)*. 2024, pp. 1–7. DOI: https://doi.org/10.1145/3649329.3657378.

[C22] Luisa Cicolini, Filippo Carloni, Marco D Santambrogio, and **Davide Conficconi**. "One Automaton To Rule Them All: Beyond Multiple Regular Expressions Execution". In: *IEEE/ACM International Symposium on Code Generation and Optimization*. 2024, pp. 1–15. DOI: https://doi.org/10.1109/CGO57630.2024.10444810.

[C21] Paolo S. Galfano, Giuseppe Sorrentino, Eleonora D'Arnese, and **Davide Conficconi**. "Co-Designing a 3D Transformation Accelerator for Versal-Based Image Registration". In: *IEEE International Conference on Computer Design*. 2024. DOI: 10.1109/ICCD63220.2024.00041.

[C20] Niccolò Nicolosi, Francesco Renato Negri, Francesco Pesce, Francesco Peverelli, **Davide Conficconi**, and Marco Domenico Santambrogio. "PSyGS Gen A Generator of Domain-Specific Architectures to Accelerate Sparse Linear System Resolution". In: *2024 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*. IEEE. 2024, pp. 41–47. DOI: 10.1109/IPDPSW63119.2024.00015. URL: https://doi.org/10.1109/IPDPSW63119.2024.00015.

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